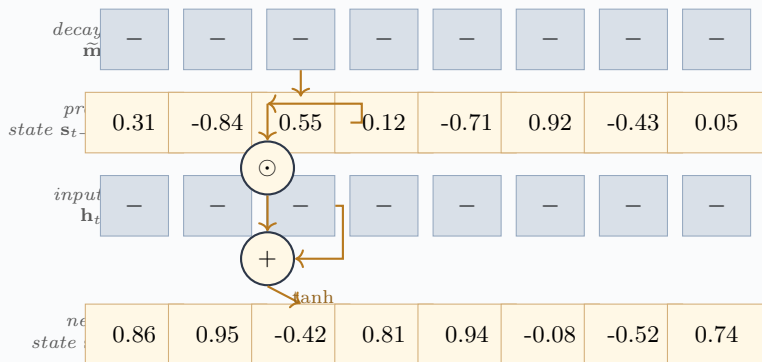


Block 3a. Recurrence — per-channel detail at one timestep t

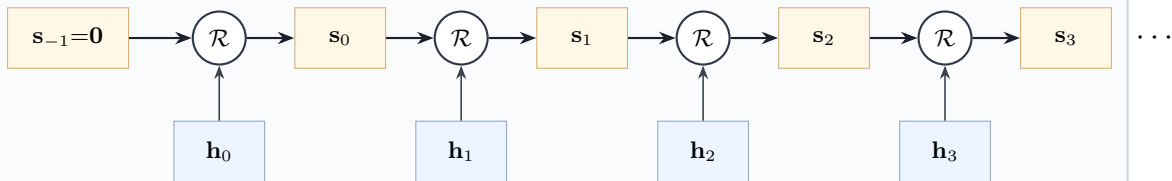


channel $i=2$:

$$\begin{aligned} \tilde{m}_2 s_{t-1,2} + h_{t,2} &= (+1)(0.55) + (-1) \\ &= -0.45 \\ \tanh(-0.45) &\approx -0.42 \end{aligned}$$

$$\mathbf{s}_t = \tanh(\tilde{\mathbf{m}} \odot \mathbf{s}_{t-1} + \mathbf{h}_t) \quad (\text{one channel at a time})$$

Block 3b. Recurrence — unrolled across time



$$\mathcal{R}(s_{t-1}, h_t) = \tanh(\tilde{\mathbf{m}} \odot s_{t-1} + h_t) \quad O(d) \text{ per step. No attention. No key-value cache.}$$